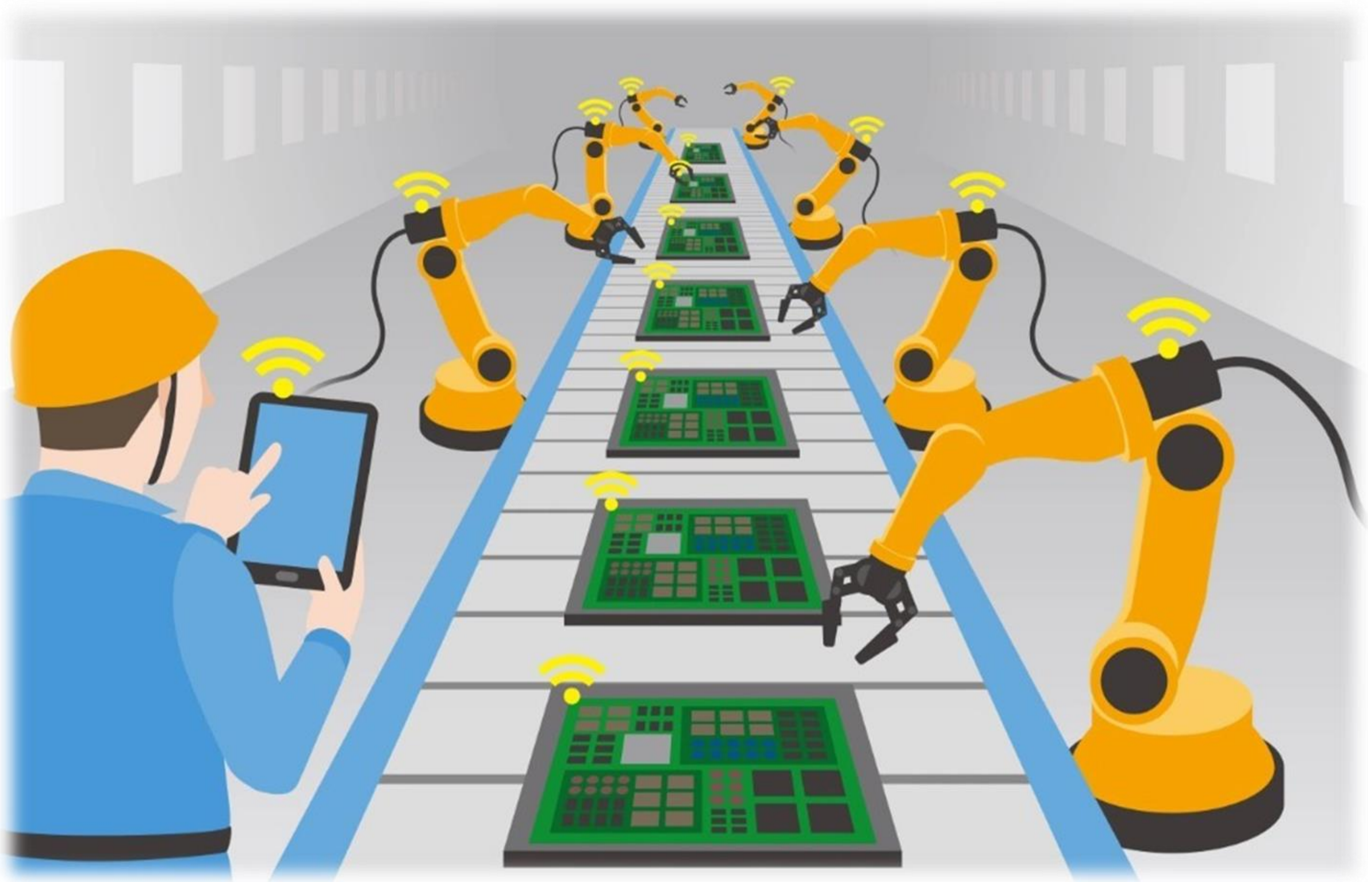
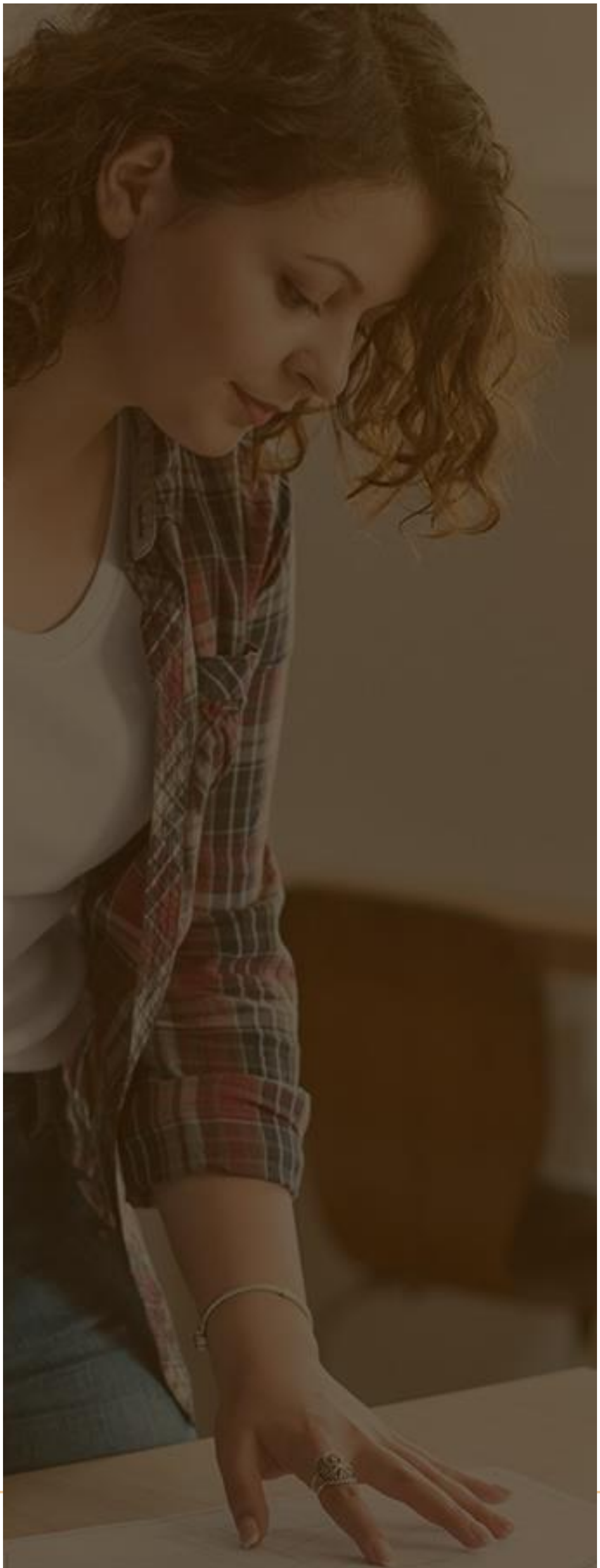


D365 BC MANUFACTURING QUESTIONS & ANSWERS



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1. PRODUCTION BOM

1. **What is a Production Bill of Materials (BOM)?**

   A Production Bill of Materials (BOM) is a comprehensive list of all the components, raw materials, subassemblies, and sometimes other BOMs, needed to manufacture a parent item. It details each component's description, the quantity required, the unit of measure, and potentially other crucial information such as routing links, scrap percentages, and valid date ranges for the component's use within the production process.

2. **What types of components can be included in a Production BOM?**

   Production BOMs can include several types of components:

   - **Purchased items** (raw materials).
   - **Produced items** (subassemblies that are also manufactured and have their own BOM).
   - **Phantom BOMs** (which are non-stocked subassemblies). A component item can be both a subassembly or a manufactured item.

3. **What are the different statuses that a Production BOM can have, and what do they mean?**

   A Production BOM can have the following statuses:

   - **New:** The BOM is newly created and is editable.
   - **Certified:** The BOM is approved for use in production planning, production orders, and cost calculations. A certified BOM cannot be edited.
   - **Under Development:** The BOM is a certified BOM that is currently undergoing edits or changes. A BOM with this status is not used in cost rollups.

  - **Closed:** The BOM is no longer in use and will not be considered in production processes.

4. **How does Microsoft Dynamics NAV handle changes to Production BOMs?**

   Dynamics NAV provides several ways to handle changes:

   - **Direct changes to a BOM**: If changes are not tracked, you can directly modify the BOM while in "New" or "Under Development" status.
   - **Creating BOM Versions**: If change tracking is required, new versions of a BOM can be created. Each version has a starting date and only one version is active on a given date, allowing for the management of engineering changes over time.
   - **Starting and Ending Dates:** You can specify start and end dates for component lines within a BOM. This enables component changes to occur automatically on specific dates, which is useful for temporary substitutions or promotions.
   - **Batch Jobs:** The 'Exchange Production BOM Item' batch job allows mass replacement of components. The 'Delete Expired Components' job can remove obsolete components.

5. **What is Phantom BOM, and why would I use it?**

   A phantom BOM is a type of production BOM that represents a non-stocked subassembly. It's a group of items that are assembled immediately before their use in a parent BOM. Phantom BOMs simplify BOM structures and streamline the MRP process by allowing planners to consider components without creating separate item cards or production orders for that subassembly. The time required for assembling a phantom BOM is considered to be zero or accounted for in the master item's routing.

6. **How can Calculation Formulas be used in Production BOMs?**

   Calculation formulas within BOM lines enable dynamic quantity calculations for components. Rather than having a fixed quantity, the required amount of a component can be determined by multiplying the "Quantity per" value by other properties such as length, width, depth or weight. The system supports formulas like *Length, Length * Width, Length * Width * Depth, and Weight, allowing for more accurate consumption tracking, especially for materials with variable shapes or sizes.

7. **What is the "Where-Used" feature and how can it be helpful?**

   The "Where-Used" feature allows users to trace the usage of a specific component or a production BOM within the product structure. This feature helps find which parent items use specific components. It is useful for tracking the impact of changes on components and production BOMs. It's available at both the BOM and component level and is used for tracing items in recalls, or determining the impact of ECOs.

8. **What reports are available for analyzing Production BOMs?**

   Microsoft Dynamics NAV offers several Production BOM reports:

   - **Quantity Explosion of BOM:** This report provides a detailed, indented list of components needed for a parent item, down through all levels of the BOM hierarchy.
   - **Where-Used (Top Level):** This report traces a component's usage upward within the product structure, showing where a specified item is utilized as a component in other parent items.
   - **Compare List:** This report allows users to compare the components of two different items, showing differences in components, costs, quantities, and cost shares.

2. BASIC CAPACITIES AND ROUTINGS

1. **What are the three hierarchical types of capacities that can be defined in a manufacturing environment?**

   The three types of capacities are:

   - **Work Center Groups**, which represent broad departments.
   - **Work Centers**, which are specific locations where work is performed such as machines or groups of people; and
   - **Machine Centers**, which are the lowest level, representing individual machines or people and often used for resources that need to be monitored closely, or which frequently cause bottlenecks.

   These are arranged hierarchically, with machine centers assigned to work centers and work centers assigned to work center groups.

2. **What is routing, and why is it important in manufacturing?**

   A routing specifies the *sequence of operations* (steps) required to manufacture a product. It is critical for production process scheduling, capacity planning, and the creation of manufacturing documents. Routings also define where each step of the production process is carried out, either at a work center or a machine center.

3. **What elements can be defined within the capacity setup?**

   Capacity setup includes several elements that describe the manufacturing environment: capacity unit of measure (like hours or minutes), work shifts, shop calendars, work center groups, work centers, machine centers, standard tasks, stop codes, scrap codes, and capacity-constrained resources. These elements are crucial for tracking and planning production capacity effectively.

4. **What are the two categories of throughput time, and what components make up each?**

   Throughput time is divided into two categories:

   Productive Time, which includes setup time (time to prepare a machine for a production run) and run time (time to actually process or manufacture an item); and

   Non-Productive Time, which includes wait time (time between process steps), move time (time a lot spends in transit), and queue time (time a lot waits at a work center before processing begins). Understanding and managing these different types of time is key to optimizing production schedules.

5. **What are routing link codes, and how are they used?**

   Routing link codes are used to *connect components on a Bill of Materials (BOM) to specific operations within a routing*. This allows for precise control over when materials are flushed or removed from inventory, based on the start date of the operation needing the materials and the component's flushing method (manual, forward, or backward).

6. **What does "capacity constrained resource" mean, and how does it impact production planning?**

   A capacity-constrained resource is a machine center or work center identified by production planners as potentially becoming overloaded. By designating a resource as capacity-constrained, it limits how much work can be assigned to it, ensuring that the resource is not loaded beyond a specific critical level, essentially implementing a finite loading approach to avoid bottlenecks and ensure the feasibility of the schedule.

7. **What are standard tasks, and how can they benefit the creation of routings?**

   Standard tasks are predefined descriptions of common operations that include task details, tool and personnel requirements, and quality measures. Using standard tasks simplifies the process of creating routings by automatically copying all of these associated definitions to a routing line, saving time and ensuring consistency.

8. **How can routing versions be used to manage changes and track historical data in manufacturing?**

   Routing versions allow for multiple versions of a routing to be maintained, each identified by a version code and a starting date. This enables manufacturers to schedule the implementation of new changes to the routing without needing to change product structures and provides a way to track the history of changes to a routing, allowing evaluation of the impact of changes, and also allowing manufacturers to return to previous configurations.

3. PRODUCTION ORDERS

1. **What are the different statuses a production order can have in Microsoft Dynamics NAV, and what does each signify?**

   Production orders in Microsoft Dynamics NAV can have five distinct statuses:

   Simulated, *Planned*, *Firm Planned*, *Released*, and *Finished*.

   - **Simulated (SPO)** orders are primarily used for costing and quoting purposes and do not impact actual production or planning processes. They don't consume materials or capacities and cannot be used for manufacturing.
   - **Planned (PPO)** orders indicate an expected production, often suggested by the planning system based on replenishment needs and reorder policies. These are not a firm commitment to produce and can be deleted or replaced by the planning system. They do, however, provide input to capacity and material requirements planning.
   - **Firm Planned (FPPO)** orders are a step closer to production, acting as a placeholder in the schedule. They are generally manually created or generated from sales orders when future capacity and material requirements are known. The planning system does not delete them in later processes, though it can suggest changes.
   - **Released (RPO)** orders signify that the production order has been released for manufacturing. Once released, you can post produced quantities and component consumption against it, though this does not necessarily mean that manufacturing has actually started. The planning system does not automatically delete released orders, but can suggest changes.

- **Finished (FPO)** orders represent closed orders, indicating that manufacturing is complete or the order has been terminated. These orders cannot be changed, posted, or deleted and are used for costing, tracking, and historical reporting.

2. **What are the "Source Types" available for a production order and how do they differ?**

   There are three source types for production orders: *Item*, *Family*, and *Sales Header*.

- **Item** source type is used when you are creating a production order for a single inventory item. If the production order is connected to a sales order, there is one production order created for each sales order line.
- **Family** is used when a group of items needs to be manufactured together, representing a collective need.
- **Sales Header** is used to create one production order for all items in a sales order, making it useful in large projects with long lead times.

3. **What is the purpose of the "Refresh Production Order" function, and how does it affect the schedule and components of a production order?**

   The "Refresh Production Order" function is used to schedule or reschedule a production order. It also recalculates and updates the order lines, routing operations, and component needs based on the parameters defined. This batch job includes simultaneous planning and scheduling of the components and routing.

   When calculating, it can perform forward or backward scheduling. ***Forward scheduling*** starts from a defined date/time and calculates the order forward. ***Backward scheduling*** starts from an end date/time and calculates the required start date/time.

   It is also responsible for generating or replacing production order lines, routing operations, and components based on the active production bill of materials (BOM) and routings associated with the item. This function will also adjust the due date of a production order as needed. It should be used when a quantity change has been made to the order header. This function cannot replace production order lines, routings, or components if ledger entries or reservations exist against the order.

4. **How does Microsoft Dynamics NAV handle the scheduling of routing operations for a production order?**

   The scheduling of routing operations involves various time elements, including **setup time, run time, wait time, and move time, but *not* queue time**. When **forward scheduling**, the program calculates from the starting date of an operation to the end date. With **backward scheduling**, the program works in reverse, starting from the required end date to determine the required start date. The system uses the **work/machine center calendars** to calculate the schedule based on operation times. The ending date/time for the final operation will determine the ending date/time for the order line. If this is later than the due date minus one day, the system will adjust the due date.

5. **How are components scheduled in a production order, and what are "Routing Link Codes"?**

   In its simplest form, component scheduling is based on the lead times for purchased components and the routing schedules for produced components. By default, components are scheduled to be available at the start of the production order.

   Routing Link Codes are used to link a component on a production BOM to a specific operation in a production routing. This allows consumption of the component to be recorded at the time the operation occurs, not just at the start or finish of a production order.

   They support just-in-time (JIT) functionality and are frequently used in long production processes.

6. **What types of changes can be made to a production order, and what considerations are necessary?**

   You can change many elements of a production order, including:

- **Production order status:** Moving it through different phases (Simulated, Planned, Firm Planned, Released, and Finished) as production progresses. There are some restrictions on changing statuses, such as not being able to switch from Simulated to Planned or vice versa, and only being able to mark a released order as Finished.

- **Date and time values:** Changes in the due date trigger automatic recalculations of the starting and ending dates/times of the order and its components, ensuring consistency.

- **Quantities:** Adjustments to the quantity field in the order header require use of the Refresh **Production Order function** to update order lines, routings, and components. Changes to the quantity field in an order line will recalculate the quantities for the components and reschedule the routing lines.

- **Routing lines:** You can modify a production order's routing, for example, to use a different machine center, particularly in make-to-order scenarios.

- **Component lines:** Components can be added, deleted, or modified, especially in MTO environments.

- **Item substitutions:** You can substitute an item on a production order component line with a previously defined substitute.

7. **What is the purpose of the "Replan" function on a production order, and how does it differ from the planning worksheet or order planning tools?**

   The **Replan** function is a tool to create and replan production orders for lower-level assemblies that are required by the current (parent) production order. It does not include time phasing, so it should not be used as a substitute for more advanced tools. It differs from the planning worksheet, which has a full planning capability, and order planning, which is a simpler manual tool. The "Replan" function is useful for multilevel production orders where the parent and subassemblies are all part of the same production order. The "Replan" function can replan at different levels: No Levels, One Level, or All Levels. No Levels will not create new production orders for lower-level items. One Level will create production orders one level down the BOM. All levels will create production orders for all levels of the BOM.

8. **What are "Phantom BOMs" and how are they treated in Microsoft Dynamics NAV production orders?**

   A **phantom BOM** is a bill of materials for a non-stocked subassembly. It simplifies product structure by allowing the MRP process to account for the components without having to manage an item card for the subassembly. When a production order with a phantom BOM is refreshed, the phantom BOM is "exploded," meaning that its components are pulled into the production order directly, rather than treating the phantom as a subassembly to be produced separately. The components of the phantom become level one components of the production order.

4. PRODUCTION ORDER PROCESSING

1. **What are the primary types of data recorded during the execution of a production order?**

   During production order execution, three main categories of data are recorded:

   - The **materials picked or consumed** for the order,
   - the **time spent** working on the order, and
   - the **quantity of the manufactured item** that is produced.

   These values are critical for tracking costs and progress throughout the manufacturing process.

2. **What is 'flushing' in the context of production orders, and what are its different methods?**

   Flushing refers to the process of recording material consumption, time spent, and production output. It can be done automatically or manually.

   The main methods are:

   - **Forward flushing**: Where material consumption occurs at the start of the order or operation.
   - **Backward flushing**: Where consumption happens when the order or operation is finished.
   - **Manual flushing**: Where recording is done by user entry.

3. **What is forward flushing and when would it be used?**

   Forward flushing is when component materials are automatically consumed when a production order is released or refreshed.

   This method is based on the expected output quantity multiplied by the required component quantities and is suitable for production environments that have few defects, few operations, and high component usage in early operations. It can be set to happen for the entire order or by specific routing operations.

4. **What is backward flushing and in what situation is it appropriate?**

   Backward flushing is when materials are automatically consumed when a production order is changed to a "Finished" status.

   Consumption quantities are based on the actual output quantity of the manufactured item. Backward flushing is often used in scenarios where it is more accurate to determine material usage based on actual completed production, not on anticipated usage. Like forward flushing, it can be set at the order or operation level.

5. **How does manual flushing work, and why might it be preferred?**

   Manual flushing involves recording material consumption and production output directly by using journals, like the ***consumption journal*** or the ***output journal***.

   This method is appropriate when there are frequent component substitutions, unexpected scrap, or when you need a more granular level of control. You can record consumption based on expected or actual output.

6. **What is the purpose of the consumption journal and how does it interact with warehouse management systems?**

   The **consumption journal** is used to *manually record and post material consumption for a* ***production order***.

When using bin functionality, the consumption journal can generate precise pick lists, specifying the bin locations of each component. This facilitates efficient material picking and can integrate with warehouse systems if desired for detailed tracking. You can also use this journal to reverse entries.

7. **What is the role of the output journal, and what does it track?**

   The **output journal** is used for recording the *production output of finished goods and the time spent on production order operations*.

   You can track setup time and run time, as well as the quantity of items produced, and scrap. Posting to the output journal will increase inventory, while also creating capacity ledger entries to track time spent. You can reverse these entries using the output journal as well.

8. **What is the production journal, and how does it differ from the consumption and output journals?**

   The **production journal** combines the functions of the **consumption** and **output** journals into a single interface. It allows users to *register consumption and output for a production order from one location*.

   It shows components and operations together with their related information, offers the ability to filter by flushing method, shows quantities already posted, and it is directly accessible from the production order. It is meant to provide a more streamlined process for those who want to combine the functions of both journals.

5. IMPORTANT MANUFACTURING SETUPS

1. **What is the purpose of the Manufacturing Setup page in Dynamics NAV, and what are some key settings that can be configured there?**

   The Manufacturing Setup page allows you to define how your manufacturing processes are managed within D365 BC.

   Key settings include:

   - Defining Work Day start and end times.
   - Setting how the Output Quantity field in production journals is populated (e.g., automatically to expected or zero values).
   - Specifying the default capacity unit of measure for production resources.
   - Setting planning warnings.
   - Managing how document numbers are assigned to production orders.
   - Specifying forecasts and managing dampener periods and quantities that affect rescheduling.
   - You can also choose whether setup times should be included in standard cost calculations.

   This page provides essential global parameters that govern the entire manufacturing process.

2. **How does the "Preset Output Quantity" setting in Manufacturing Setup impact production journals?**

   The **Preset Output Quantity** setting determines how the **Output Quantity** field is automatically populated when a **production journal** is opened.

You have three options:

Expected Quantity: This fills the field with the planned production order quantity.

Zero on All Operations: This sets the field to zero for all operations.

Zero on Last Operation: This sets it to zero only for the final operation.

This setting can streamline data entry by pre-populating the output quantities, reducing the need for manual adjustments.

3. **What is the difference between Make-to-Stock (MTS) and Make-to-Order (MTO) manufacturing policies, and how do they affect production orders?**

   Make-to-Stock (MTS) items are produced for inventory based on anticipated demand, typically standard items with shorter lead times. The planning system in MTS considers only the first level of the BOM (Bill of Materials) and typically creates one production order per item.

   Make-to-Order (MTO) items are produced specifically for a customer order, often for complex and/or expensive items. The planning system explodes the entire BOM for MTO items, potentially creating multiple production orders at various levels to fulfill the customer order. MTO items also establish reservations between demand and supply, preserving order-specific customizations. If you want a multilevel production order all levels must be set to MTO.

4. **How does the "Dampener Period" and "Dampener Quantity" impact rescheduling and quantity change suggestions from the planning system?**

   The **Dampener Period** prevents rescheduling a supply order to a *later date* if the proposed date is within a defined period. It only considers rescheduling to a later date if the new proposed date exceeds the existing order's due date plus the dampener period. Rescheduling to an earlier date is always allowed.

   The **Dampener Quantity** blocks insignificant quantity *reduction* suggestions for existing supply orders. If a suggested reduction is less than the dampener quantity, it won't be proposed, but suggested increases are not impacted.

These settings help avoid constant minor adjustments to orders. Dampener settings on the item card take precedence over the default values set in the Manufacturing Setup.

5. **What is the role of the "Reordering Policy" field on the item card and what are the differences between *Fixed Reorder Qty*, *Maximum Qty*, *Order*, and *Lot-for-Lot*?**

   The **Reordering Policy** dictates how the planning system calculates lot sizes and replenishment timing.

   - **Fixed Reorder Qty.** uses a predefined quantity (Reorder Quantity) as the standard lot size, typically used with a reorder point.
   - **Maximum Qty.** uses the Maximum Inventory level to calculate lot sizes, again used with a reorder point.
   - **Order** generates a unique order for every demand and automatically reserves the supply to the demand.
   - **Lot-for-Lot** aggregates all demands within a specific planning period and creates one supply order to fulfill them.
   - **Blank** option means no automatic order proposals are created.

   The choice of reordering policy should align with item characteristics, demand patterns and business strategy.

6. **How do safety stock, safety lead time, reorder point, and reorder quantity interact to ensure sufficient inventory levels in D365 BC?**

   - **Safety Stock Quantity** is the minimum acceptable inventory level, designed as a buffer against demand fluctuations. The planning system does not include it in any calculations, it only checks against it.
   - **Safety Lead Time** provides a buffer time for potential delays, added to both forward and backward scheduling calculations to determine when manufactured items need to be available.

- **Reorder Point** is an inventory threshold below which replenishment is triggered. The system creates a supply order to bring stock back up to at least this point. The reorder point must always be set above the safety stock quantity.

- **Reorder Quantity** specifies a standard lot size for replenishing the item. Only the "Fixed Reorder Quantity" policy uses this field. The program uses this as a minimum, though it may be increased. These parameters work together to balance inventory levels, prevent shortages, and optimize production planning.

7. **What is the purpose of the Stockkeeping Unit card, and what are some of its unique fields related to manufacturing?**

   The **Stockkeeping Unit (SKU)** card allows you to configure item properties at the location level, overriding the item card's settings.

   This is especially useful when different locations have different stock levels or lead time constraints. Unique fields include: different replenishment systems per location, a transfer-from code for transfers to this location, and a "Components at Location" field which specifies where component items are supplied from for production when the item is produced at a specific location, overriding both the item card's settings and default setups. Using SKU cards with mandatory locations enabled is essential for accurate planning and production across your company.

6. FORECASTING AND PLANNING

1. **What is the fundamental difference between independent and dependent demand in the context of production planning?**

    - **Independent demand** refers to the demand for finished goods that are directly sold to customers. Examples include bicycles or replacement parts.
    - **Dependent demand**, on the other hand, is the demand for raw materials, components, or sub-assemblies that are necessary to produce those finished goods. For instance, the demand for bicycle wheels and bells are dependent on the demand for finished bicycles.

2. **What are the three available forecast types in D365 BC, and how do they relate to different types of demand?**

    The three forecast types are:

    - **Sales Item:** *Sales Item* is used to forecast the independent demand for finished products.
    - **Component:** *Component* is used to forecast the dependent demand for items needed in the production of finished goods.
    - **Both:** The *Both* option is for viewing combined totals of sales items and components but cannot be used to enter data.

3. **When entering forecast data for a period such as a month or a week, on what date does the system record the forecast entry, and how does this impact planning calculations?**

    When entering a forecast value for a period other than a day, the system automatically records the entry on the first day of that period (e.g., if you enter a monthly forecast, it's recorded on the first day of that month).

By storing forecast values at the day level, the system can aggregate these values to view or calculate for other time periods like weeks, quarters, or years.

4. **What are the roles of Master Production Scheduling (MPS) and Material Requirements Planning (MRP), and how do they interact with production forecasts?**

   MPS is used to create a master production schedule for end items based on actual demand and the production forecast. It's for items with a forecast or sales order line.

   MRP, in turn, calculates the material requirements for components needed to fulfill the MPS. It is calculated only for items that are not MPS items.

   They are both used to net demand against supply and suggest actions for any necessary changes. The planning system uses the same planning algorithms for both MPS and MRP and can be run separately or together.

5. **Explain how sales orders impact a production forecast and provide an example of this interplay.**

   Sales orders reduce the impact of the production forecast.

   For example, if you forecast 200 units of a product and a sales order for 50 units of that product is created, then the remaining forecasted demand is reduced to 150 units.

   The total demand would remain at 200 (150 forecasted and 50 actual sales). This is a process called "**netting**" where actual sales orders "**consume**" the corresponding forecasted demand.

6. **What is the role of order tracking in relation to production forecasts, and what are its limitations when dealing with forecasts?**

   Order tracking provides a connection between supply and its corresponding demand. It explicitly traces a planning line to a specific sales order.

   However, order tracking cannot track back to a production forecast directly because of the way forecasts are designed. A forecast quantity is often a sum of several entries, and the tracking system needs to trace to a specific demand line. Forecasts do not represent a singular demand line, as a sales order might, thus limiting tracking ability.

7. **What are low-level codes and why are they important in planning?**

   Low-level codes represent the position of an item within a Bill of Materials (BOM) hierarchy.

   The finished good is at level zero, components are level one, sub-components level two, and so on. Accurate low-level codes are crucial because the planning system processes items in order of these codes (starting at level zero). If they're incorrect, it can lead to inaccurate action messages.

   These codes are calculated when you certify a production BOM attached to an item card (if **Dynamic Low-level Code** is selected), or when you run a batch job.

8. **What is the Planning Worksheet and how is it used?**

   The **Planning Worksheet** is where D365 BC displays action messages, which are suggestions to create new supply or change existing supply. These suggestions result from the planning calculations (MPS and/or MRP) and aim to balance supply with demand.

   Action messages are for planned or firm production orders, assembly orders, purchase orders, and transfer orders. The production planner can review, modify and accept these action messages before they are carried out by the system. The worksheet acts as a central hub for managing the production plan.

9. **What are the key differences between Regenerative and Net Change Planning?**

   Regenerative planning recalculates the plan for all selected items, regardless of any changes since the last planning run. It is typically used when significant changes to master data or capacity occur.

   Net change planning only focuses on items that have experienced changes in demand, master data or supply orders since the last planning run. This method is faster and uses fewer system resources because it processes fewer items, focusing on only the affected areas.

While both methods will generally create the same supply suggestions for the items that have been affected, **net change** is ideal for daily planning and **regenerative** for less frequent, overall planning changes.

10. **What are the various "Action Messages" that the planning system can generate, and what do they mean?**

The five action messages the system can generate are:

- **New:** Suggests creating a new supply order (production, purchase, etc.) to cover uncovered demand.
- **Change Qty.:** Suggests changing the quantity of an existing supply order to match changed demand.
- **Reschedule:** Suggests rescheduling the due date of an existing supply order to match changes in demand date.
- **Resched. & Change Qty.:** Suggests changing both the due date and quantity of an existing supply order due to changes in demand.
- **Cancel:** Suggests cancelling an existing supply order when related demand is removed.

11. **What are item variants and how are they used in planning?**

Item variants represent different characteristics of an item, such as size or color. They allow you to manage variations of the same item without creating entirely new item records.

In planning, item variants are treated as distinct items, participating only if there is specific demand for that variant.

For instance, a bicycle can have variants for different frame sizes. These variants share the same Bill of Materials (BOM) and routing as the parent item, simplifying maintenance and ensuring consistency across all variants.

12. **What are Stockkeeping Units (SKUs) and why are they important for planning in multiple locations?**

    A **Stockkeeping Unit (SKU)** is an extension of an item card that defines inventory and planning parameters for an item or item variant at a specific location.

    SKUs are essential for managing planning in multi-location environments because they enable you to set location-specific replenishment methods, component locations, and planning parameters. Without SKUs, the system uses the item card's default information.

    This functionality allows, for instance, manufacturing an item at a different location than where it is sold, by setting a transfer order from a manufacturing location to the selling location.

13. **How can you change the replenishment system for a planning line, and what does refreshing planning lines mean?**

    The replenishment system on a planning line can be changed, for example, from “Prod. Order” to "Purchase." This allows you to make changes to how the supply is created when you are carrying out action messages.

    For example, if there is a capacity constraint, you can shift production to purchasing.

    You can use the "Refresh Planning Line" function to update a planning line. It deletes existing component and routing information and generates new component and routing lines based on the standard BOM and routing data. This is useful when there are updates to a BOM or routing that you need to take into account.

7. SUBCONTRACTING

1. **How do you identify a work center as a subcontractor's work center within Microsoft Dynamics NAV?**

   To designate a work center as belonging to a subcontractor, you must specify a **vendor** in the **Subcontractor No.** field on the work center card.

   This field is crucial for the system to recognize that this work center is used for subcontracted operations.

2. **How does costing work for subcontracted operations, and how can you specify different costs for different processes with the same vendor?**

   You can set up costing for subcontracting work either at the work center level or on individual routing operations. To use a single rate for a subcontractor across all processes, you set up costs on the work center card and leave the **Specific Unit Cost** field blank.

   However, if you need to have different costs for each process with the same vendor, select the **Specific Unit Cost** field. You can then specify costs in the **Unit Cost per** field on each routing line assigned to the subcontractor work center. This approach allows for flexible cost management based on the specifics of each subcontracted operation.

3. **What is the Subcontracting Worksheet and how does it work?**

   The **Subcontracting Worksheet** is a tool used to manage subcontracting activities. It calculates production order routing operations scheduled for subcontractor work centers and then generates action messages.

   These messages suggest new purchase orders to be sent to the subcontracting vendors. This tool allows you to review and modify actions before they are carried out. It is similar to the planning and requisition worksheets in its approach. The worksheet primarily focuses on creating purchase orders and does not have a replenishment system.

4. **How are purchase orders generated for subcontracted operations in D365 BC?**

   Purchase orders are generated for subcontracted operations via the **Subcontracting Worksheet.**

   The worksheet's "**Calculate Subcontracts**" function identifies released production orders with routing operations assigned to a subcontractor's work center. It then proposes new purchase orders based on these operations. Each action message in the worksheet can be individually accepted or rejected before creating the purchase order.

5. **How does posting a subcontracting purchase order impact a production order?**

   Posting a subcontract purchase order impacts the production order primarily by registering the costs of the subcontracted service.

   Receiving a subcontract purchase order only increases the output quantity of the production order's parent item if the subcontracted operation is the *last* operation on the production order routing. In all other cases it does not affect the output quantity. However, when the purchase order is invoiced, the costs from that purchase order flow through to the production order.

6. **What is the importance of linking purchase orders to production orders in the context of subcontracting?**

   Linking purchase orders to the source production order operation ensures **accurate costing** and **prevents duplicate purchase orders** from being proposed in future planning runs.

   If the cost of the subcontracted service changes when you invoice the purchase order, these updated costs will correctly flow back to the production order. This link is critical for proper cost management.

7. **What is the purpose of the Subcontractor – Dispatch List report, and what information does it provide?**

   The **Subcontractor – Dispatch List report** provides a list of all released production order routing lines for subcontracting operations.

   It shows the scheduled dates for these operations, as well as whether a corresponding purchase order exists for each. The report allows you to filter results based on Vendor, Work Center, and Prod. Order Routing Line data. It gives a comprehensive overview of subcontracting activities, including relevant dates, quantities, and linked purchase order information, organized by

8. ADVANCE CAPACITY

1. **How do shop calendars and capacity calendars work, and what's their relationship?**

   Shop Calendars define the general working days, times, and holidays for work shifts. You can create multiple shop calendars to account for different shift schedules.

   Capacity Calendars are specific to a work center or machine center. They are derived from a shop calendar but incorporate the specific capacity, efficiency, and absences of a work center or machine center. This allows for a more accurate view of available capacity. The program uses the efficiency and capacity settings of a work center or machine center to adjust available hours, in addition to using Shop Calendar holidays. The capacity calendar is used to schedule production orders.

2. **What does the "Consolidated Calendar" option on a Work Center card mean, and how does it affect capacity planning?**

   When the "**Consolidated Calendar**" option is selected for a Work Center, the work center itself does not have its own capacity. Instead, its capacity is calculated as the *sum* of the capacities of all the Machine Centers assigned to it.

   This setup is ideal if you are scheduling production at the machine center level. If not selected, the work center's capacity will be based on its own capacity setting. It allows you to view aggregate load when you consolidate the calendar.

3. **How can you account for time when a work center or machine center is not available for production?**

   There are a few methods to account for **non-production time**.

- **Shop Calendar Holidays**: Shop Celender Holidays can be defined to remove capacity from all facilities that use a particular calendar.

- **Planned Absences:** You can register *planned absences* directly against work centers or machine centers, indicating downtime for specific periods. The system provides options to register absences manually, via the "Register Absences" page, or using batch jobs to update multiple facilities at once.

When absences are registered, the capacity calendar for that work center or machine center will be reduced.

4. **What is the purpose of a Capacity Journal, and when should it be used?**

   The **Capacity Journal** is used to directly post time consumption to work centers or machine centers.

   You can use it to record maintenance work, track stop time, or log any other time that consumes capacity. This journal is distinct from output and production journals, which are used to record time associated with production orders. The Capacity Journal posts directly to the facility (work center or machine center) and does not reference a production order.

5. **What are some of the key fields on a Work Center or Machine Center card that impact capacity planning?**

   Key fields include:

   - **Capacity:** The amount of work that a facility can finish in a specified time.
   - **Efficiency:** The output of a facility in relation to the expected standard output (expressed as a percentage).
   - **Unit of Measure Code:** The unit used to calculate capacity and costs (minutes, hours, days, etc.).
   - **Shop Calendar Code:** The Shop Calendar used to define working hours for the facility.
   - **Direct Unit Cost:** The direct labor rate for the facility per unit of measure.

- **Indirect Cost %:** The indirect costs as a percentage of direct unit costs.
- **Overhead Rate:** An absolute amount to cover other expenses of the facility, such as maintenance.
- **Setup Time:** Time needed to set up the machine.
- **Wait Time:** Time spent waiting after completion.
- **Move Time:** Time spent in transit between operations.
- **Fixed Scrap Quantity/Scrap %:** Anticipated loss due to scrap.
- **Send-Ahead Quantity:** The minimum quantity that can be processed before the next operation starts.
- **Concurrent Capacities**: How much capacity can be planned at the same time.

6. **How do I calculate the unit cost of a work center or machine center, and how is that used?**

   Unit costs can be calculated using the formula:

   *direct unit cost + (indirect cost % * direct unit cost) + overhead rate*

7. **What happens if a work center or machine center is blocked?**

   If a work center or machine center is marked as "**Blocked**," it cannot be used for production or any posting, including time entry via the capacity journal.

   This is generally used when a facility is out of service. The setting does not change the capacity calendar but does mean no production can be posted to the blocked facility.

9. SHOP LOADING

1. **What is shop loading, and why is it important in manufacturing?**

   Shop loading refers to the process of comparing the demand for work at production facilities (work centers and machine centers) against their available capacity.

   It's crucial because it helps production planners identify potential bottlenecks, overloads where demand exceeds capacity, and underutilization, where resources are idle. Effective shop loading ensures that production schedules are realistic and that resources are used efficiently, avoiding scheduling and delivery problems.

2. **What is the difference between a work center and a machine center in the context of shop loading?**

   A ***work center*** represents a department or area where work is performed and can include multiple *machine centers*.

   A **machine center** is a specific piece of equipment or workstation within a work center.

   In the system, both work and machine centers have capacity, and both are assigned in production routings. A work center can be associated with multiple machine centers.

   The Work Center Load page displays information for the work center and all its related machine centers while the Machine Center Load page focuses on the information of a single machine center.

3. **How are the capacity and load calculated for work and machine centers?**

   Capacity is determined by the shop calendar, efficiency, and capacity values defined in the work center or machine center card. It reflects the total work that a facility can complete within a given period. Load is calculated by dividing the allocated capacity (demand from production orders) by the available capacity for a specific period. This is expressed as a percentage and shown on the load pages.

4. **What does "infinite capacity loading" mean, and what are the implications for production planning?**

   Infinite capacity loading implies that the planning system will schedule work at a production facility regardless of its capacity limitations.

   This can lead to unrealistic schedules and overloads if planners do not monitor the load and make necessary manual adjustments. It is the default behavior for the system unless finite loading is explicitly turned on for a resource. It is the production planner's responsibility to monitor load and ensure schedules are realistic when working with infinite capacity loading.

5. **What is finite capacity planning, and how does it help avoid overloads?**

   Finite capacity planning allows a planner to identify specific resources as constrained. With finite capacity planning, the system schedules work up to a specified percentage (Critical Load %) of a facility's capacity, and pushes excess work to the closest available period within its due date.

   This is particularly important when managing bottlenecks or critical resources as it ensures production schedules are realistic and avoids overloads.